

MACHINE LEARNING BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM

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MULTIMEDIA UNIVERSITY

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BY

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ABSTRACT

Malicious and phishing websites remain one of the most common and effective attack points of cyberattacks, both at the individual and organisational levels. The sites will mimic authentic Internet-based services to mislead people to reveal sensitive data or malware code that instances a breach of systems. Old methods of site protection, including browser alerts, static blacklists, and signature-based detection systems are becoming less useful as they cannot identify the newly developed or rapidly changing malicious websites. Attackers often alter URL formats, domain attributes, and content of web pages in order to avoid detection causing delayed or inefficient protection.

To overcome these shortcomings, the present project will present a machine learning-based Malicious Web Site Detection and Response System that will be able to detect malicious Web sites in real-time and implement a proper security reaction in case of threat evaluation. The suggested system will examine a set of URL lexical derivatives, domain attributes and webpage building attributes to categorise the web pages as legitimate or malicious. Several trained machine learning models are compared systematically in order to determine the most appropriate classifier in as far as detection, resistance, interpretability, and real-time implementation.

According to the classification confidence, threat assessment mechanism is a rule-based system which is used to classify the websites into various risk levels. Every level of threat is linked to a specific response measure, i.e., access permission, sending warning messages, or blocking malicious sites. This design gives a proportional response of security to the detected risk and makes the design usable and with the least user disruption.

The architecture will be designed to utilise the proxy-based system, which allows to intercept and analyse the web traffic in real-time without altering browser internals. This deployment model offers flexibility, a centralised control, and simple to integrate and thus suitable in academic research as well as real-life security prototypes. Moreover, transparency and explainability are highlighted in the project

as the focus is given to lightweight feature sets and classical machine learning models, which enable detection decisions to be interpreted and explained.

The results of this project will be a working prototype that will prove the viability of the combination of machine learning-based detection and automated response mechanisms. The work will help develop practical, efficient and interpretable web security solution, which will provide additional security against the emerging malicious websites and a solid base on future system refinement and assessment.

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LIST OF ABBREVIATIONS/ SYMBOLS

API	Application Programming Interface
CNN	Convolutional Neural Network
DOM	Document Object Model
F1	F1-Score
FN	False Negative
FP	False Positive
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ML	Machine Learning
MITM	Man-in-the-Middle
RF	Random Forest
SVM	Support Vector Machine
URL	Uniform Resource Locator

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CHAPTER 1

INTRODUCTION

1.1 Overview

Attacks through the web, especially phishing and rogue websites- are still one of the most popular and efficient initial vectors of intrusion in cybercriminals (Tian et al., 2025; Türk et al., 2025). Hackers regularly develop fake or dangerous websites to steal credentials, send malware or use browser vulnerabilities (Nowroozi et al., 2024). Conventional blacklist-based security solutions have been found to be ineffective in detecting newly developed or fast changing malicious sites because attackers frequently alter domains, URLs, and webpage layouts in order to circumvent filtering measures (Aslam et al., 2024; Haq et al., 2024).

In this project, it is suggested to develop the Malicious Web Site Detection and Response System based on the concepts of machine learning that will be capable of categorizing websites as malicious or legitimate based on a set of URL characteristics, domain data, and web page structural characteristics (including HTML, scripts, and embedded objects) (Zhang et al., 2024; Utku & Can, 2022). The system is expected to detect the threat in real-time and start the relevant response in accordance with the threat severity, which will enhance the security of the user and reduce the chances of encountering phishing or malicious materials (Guo, 2023). A viable prototype - developed as a browser extension or proxy-based filter - will help to prove that in the case of the system, it is possible to issue warnings, block harmful websites, or track events to analyse them (Nowroozi et al., 2022).

The present project is very similar to the Security Technology program because it entails the inclusion of cybersecurity notions, machine learning algorithms, and safe computing measures. It offers practical knowledge of threat detection, safe data management, and implementation of ML models in a security-oriented system (Tian et al., 2025).

1.2 Problem Statement

Phishing and rogue websites are the threats that remain very important to users, organisations, and online systems (Tian et al., 2025; Türk et al., 2025). Such sites usually resemble official services to lure users into providing sensitive data, or they focus on giving very bad scripts and payloads which are used to attack devices (Aslam et al., 2024). Current security mechanisms, including warnings on browsers, blacklists, and signature-based detection, have a weakness because they cannot easily identify zero-day sites or those that are recently developed by malicious individuals (Haq et al., 2024). The traditional techniques find it difficult to offer timely and correct protection because attackers often modify the URL structures, change the content of the web pages and employ obfuscation techniques (Nowroozi et al., 2024).

Hence, a system that can detect these risks in real time and intelligently should be developed to curb such risks (Zhang et al., 2024). Machine learning provides a possible remedy to this because it detects malicious patterns along the lines of learned characteristics, not using only predefined signatures (Utku & Can, 2022). Nevertheless, a gap exists between the practice, lightweight systems based on ML that can be directly integrated with the user browsing environment and react based on the severity of the risk (Guo, 2023). This project fills this gap because it will design a detection system that addresses the following requirements:

- Web site characteristics such as URL patterns, HTML structure and embedded scripts are analysed in order to determine whether the site is legitimate or malicious.
- Classifies the level of threat depending on the outcome of the classification.
- Takes an acceptable security action (e.g., warning, blocking, logging) to safeguard the user.

Its fundamental issue, thus, lies in the inaccessibility, inaccuracy, and responsiveness of a machine learning-based system that actively seeks out malicious sites and automatically responds to them depending on the severity of the danger (Nowroozi et

al., 2022). The given project will offer a solution to this issue based on a properly designed detection-and-response prototype (Tian et al., 2025).

1.3 Project Objectives

- To design and develop a machine learning model that identifies if a website is malicious or legitimate based on URL and the website features.
- To build a prototype system that integrates detection and provides automated responses accordingly.

1.4 Project Scope

The areas of interest in this project include the creation of a machine learning-driven system to identify malicious and phishing websites through the inspection of URL features, domain data, and webpage structural features (Tian et al., 2025; Zhang et al., 2024). This involves gathering and processing legitimate as well as malicious web sites, deriving significant features and developing suitable machine learning models to categorize the web sites in terms of these characteristic features (Haq et al., 2024; Aslam et al., 2024). Moreover, the project will aim at creating a working prototype e.g. browser extension or proxy-based filter that incorporates the trained model to make real-time detection and carry out appropriate responses, e.g. issue warnings, block hazardous pages, or record detected threats (Nowroozi et al., 2022; Guo, 2023).

The design of system architecture, workflow diagrams and documentation of the system are also covered in the project to show how the different elements interrelate with each other in the entire detection-and-response pipeline (Türk et al., 2025). And the performance analysis will be used to evaluate the accuracy, false-positive, and false-negative rates, and the total reliability of the system in the real-world conditions (Utku & Can, 2022). The project will however not entail the implementation of live malware or malicious scripts, it will not go to the extent of implementing the project at an enterprise level, deep malware behavioural analysis and threat identification out of the malicious websites' context (Nowroozi et al., 2024). The project is still aimed at developing an accessible, efficient and practical solution that can be applied to the end-user protection in a controlled and academic environment (Tian et al., 2025).

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Phishing and malicious websites remain one of the most popular initial attack vectors by cybercriminals because of their low cost, high success rate, and the fact that they are not reliant on technical exploitation, but through social engineering (Tian et al., 2025; Türk et al., 2025). Attacks of this nature usually deal with deceptive websites that mimic trusted services to steal credentials or provide malicious codes and sail, resulting in loss of money, identity theft and compromise of a system (Nowroozi et al., 2024).

The conventional protection systems like the static blacklist, rule-based blocking, and signature-based detection have failed to counter current web threats. New domains appear on a regular basis, URL structures are manipulated, and the content of the web pages constantly changes to avoid detection (Aslam et al., 2024; Haq et al., 2024). This has prompted the focus of the research on machine learning (ML) and deep learning (DL) methods that are able to extrapolate on known patterns and identify previously unknown malicious websites (Zhang et al., 2024).

The chapter is a literature review of the available literature on malicious websites detection basing on feature spaces and machine learning methods, vulnerability of machine learning to adversarial attacks and real-time machine learning detection and response systems. A critical and comparative analysis is done to determine the weaknesses of current work and inspire the development of the proposed system.

2.2 Feature Spaces and Data Sources

Malicious websites detection systems are highly dependent on feature selection to establish the effectiveness and efficiency of the system. The initial studies were mainly based on lexical and host-based URL attributes including URL length, special characters, suspicious tokens and domain attributes. These are lightweight in terms of computation and can be applicable to real-time detection (Utku & Can, 2022; Guo, 2023).

Nonetheless, URL only strategies are susceptible to evasion methods, since attackers can readily alter or reduce the length of URLs so that they can avoid detection. To overcome such a weakness, recent research adds the HTML and DOM structural capabilities, such as the inclusion of the forms, scripts, frames, and references to the external resources, which give more insight into the behaviour of the webpage (Zhang et al., 2024; Yoon et al., 2024).

More sophisticated research suggests the use of multi-channel feature space which is a combination of lexical, structural and visual features (e.g., webpage screenshots). These methods show better resistance to obfuscation but greatly raise the computational and inference latency (Yoon et al., 2024).

Table 2.1 Comparison of Feature Spaces Used in Prior Studies

Study	URL Features	HTML/DOM Features	Visual Features	Key Strength	Key Limitations
Utku & Can (2022)	✓	✗	✗	Fast, Lightweight	Limited Robustness
Guo (2023)	✓	✓	✗	Balanced Accuracy	Moderate Overhead
Zhang et al. (2024)	✓	✓	✓	High Detection Rate	High Computation
Yoon et al. (2024)	✓	✓	✓	Strong Against Evasion	Not Real-time

Multi-modal methods are more accurate; however, their implementation in real-time or client-side can be considered difficult. This encourages the application of

the hybrid lightweight feature sets that trade-off between detection performance and computational feasibility- a strategy followed in this project.

2.3 Machine Learning and Deep Learning Techniques

Various machine learning algorithms have been used in the malicious website detection. Random Forest, Support Vector Machine (SVM), and Gradient Boosting are considered classical supervised learning models that are easy to interpret, robust, and have a high-baseline performance when paired with engineered features (Utku and Can, 2022; Adam et al., 2024).

The automatic methods of learning the hierarchical representations of the raw URLs or webpage contents are deep learning methods, such as CNNs, LSTMs, and transformer-based methods. Such approaches tend to be more successful than classical models in case big and well-labelled datasets can be provided (Aslam et al., 2024; Haque et al., 2024). They have however, the disadvantages of bringing about cost of higher training, less interpretability and higher inference latency.

It has also been suggested that hybrid models that involve sequence learning and classical classifiers can also be used to combine the advantages of representation learning and decision stability (Aslam et al., 2024).

Table 2.2 Comparison of Detection Models

Model Type	Accuracy	Interpretability	Computational Cost	Suitability for Real-time
Random Forest	High	High		High
SVM	Medium-High	Medium	Medium	Medium
CNN/LSTM	Very High	Low	High	Low-Medium
Transformer	Very High	Very Low	Very High	Low

Despite superior performance of deep learning models, they are too complex to be deployed in real-time. The classical ML models are still the most suitable in

security systems that need transparency, low-latency, and explainability which are some of the factors given priority in this project.

2.4 Adversarial Attacks and Robustness

According to the recent research, ML-based detectors are vulnerable to adversarial attacks, such as evasion attacks, label poisoning, and feature manipulation (Nowroozi et al., 2022; Nowroozi et al., 2024). There are cases where attackers may generate inputs with a purpose of misleading classifiers or poisoning training data to worsen the performance of the model.

Some of the suggested defences are ensemble learning, data sanitization, and sound feature selection techniques that lessen reliance on the readily manipulable properties (Nowroozi et al., 2024). These results highlight the need to create systems that are resilient to adaptive adversaries besides being accurate.

The bulk of literature is on detection accuracy in an ideal setting, and little has been done on adversarial robustness in the real-world deployment. This presents a gap in light but tougher detection systems that can be applied in real-life situations.

2.5 Real-time Detection and Response Mechanisms

It is futile to detect without responding in time. Some of the studies include the idea of implementing detection models into browsers extensions or proxy-based systems to provide real-time protection (Guo, 2023; Nowroozi et al., 2022). Browser extensions give instant user feedback at the cost of browser APIs, whereas proxy-based systems can be enforced centrally and can be logged in detail. Reduced latency and features extraction through lightweight models and efficient algorithms is essential to ensure the model is usable (Utku and Can, 2022; Swetha et al., 2024).

Table 2.3 Comparison of Deployment Approaches

Deployment Method	Advantages	Limitations
Browser Extension	Immediate user feedback	API Constraints

Proxy-based System	Centralised control and scalable	Requires configuration
Server-side Filter	Powerful analysis	Higher latency

Proxy-based deployment offers a low-overhead, real time detection, and balance, and thus is appropriate to academic prototypes and security research.

2.6 Research Gap and Motivation

According to the literature, the current research pays too much attention to detection accuracy but does not consider the constraints of deployment or provides complex multi-modal approaches that are not applicable in the real-time. Lightweight, explainable and built-in detection-and-response systems that strike a balance between accuracy, efficiency and usability do not exist.

This gap is met by this project which is a combination of:

- Minimalistic feature engineering
- Classical model comparison machine learning
- Proxy-based real time deployment
- Automated responses on the threat level

CHAPTER 3

Methodology

In this project, the methodology imposed combines machine learning, analytical studies, experimentations and system development in the process of developing this malicious website detection and response system project. The methodologies used are organised as below to provide a clearer view and understanding to the development of this project.

3.1 Literature Reviews and Analytics

The initial approach to this literature-based analytical methodology is utilised to gain a better idea of the present situation in malicious websites and phishing detection practices. The literature includes academic journals, conference papers and industry reports which are reviewed to understand common attack tactics, the features of rogue websites and drawbacks of the current detection systems like blacklists and signature-based systems.

This step aims at the discussion of the previous research on the URL-based detection, webpage content analysis, and using machine learning models to ensure cybersecurity. This review is applied to the findings to determine effective sets of features, popular classifiers, performance measurements, and gaps in research. This approach guarantees that the proposed system will be constructed based on the existing knowledge as well as dealing with the current limitations.

According to the results of the literature analysis conducted in Chapter 2, this project will use a lightweight and pragmatic detection strategy that is accurate, interpretable, and performs in real-time. According to previous research, deep learning and multi-modal methods have high detection rates, but they incur high computational costs and are not transparent enough to be deployed in real-time. Thus, the project is centred around feature space and machine learning methods which have proven to be

very good and yet accessible to be implemented in either a client-sided environment or a proxy environment.

3.2 Data Collection

The data collection methodology is a literature review that relies on the available data in the published literature. An empirical approach is used in gathering the datasets to be used to train and evaluate the machine learning models. Published datasets of labelled malicious and legitimate sites are obtained on the reliable platforms that include Kaggle and phishing intelligence repositories.

The information contained in the datasets comprises URLs, domain attributes, and web page related information. It responds to academic and legal conditions by using only ethically sourced and publicly available datasets. Data collected is vetted to ascertain the quality of data, its completeness and relevancy to the project goals. The selection of publicly available URL and webpage datasets is in line with the previous literature that focuses on the efficacy of supervised learning with labelled phishing and benign websites data to detect malicious websites.

3.3 Feature Engineering and Data Preprocessing

After gathering them, the data sets are pre-processed ready to be utilised in machine learning activities. The methodology involves eliminating duplicates, the processing of missing values and the normalization of numerical features. The URLs and webpage content are then fed through the program to generate any meaningful features which can assist in differentiating between the malicious and the legitimate websites.

The features targeted by feature engineering include lexical features of the URL (i.e. URL length, special characters, suspicious keywords), domain features, and the selected webpage structural features including the existence of scripts, forms, and use of framed elements. It is based on first converting raw data on the web into structured feature vectors which can be subjected to machine learning classification.

Similar to other literature review, lexical URL features and the chosen webpage structural features are more important than the computationally expensive visual or behavioural features in this project. It has been found that the accuracy of detection and system latency is a favourable trade-off in such lightweight feature sets and are therefore suitable in real time deployment situations.

3.4 Machine Learning Model Development

The built classification models using a supervised machine learning approach are used to determine the presence or absence of malicious and legitimate websites. The ready dataset is separated into a training and a testing set, which will enable the objective assessment of the performance. In line with the results of Chapter 2, this project will involve traditional supervised machine learning models as opposed to deep learning models. Although the deep learning methods have been shown to be highly efficient in particular studies, the increased computational complexity and their low interpretability represent a limitation in security systems that are real-time. In this way, classical models are chosen due to their accuracy, efficiency and explainability.

Various machine learning methods are investigated including the Random Forest, Support Vector Machine (SVM) and gradient boosting). Practices Model training This is the process of tuning and optimization of parameters to maximize classification accuracy and minimize false-positive rates. The methods of cross-validation are employed to increase the robustness of the models and avoid overfitting. Empirical selection of models is done by carrying out a comparative analysis of various models to make sure that they are selected based on their empirical performance instead of the preference of the algorithm.

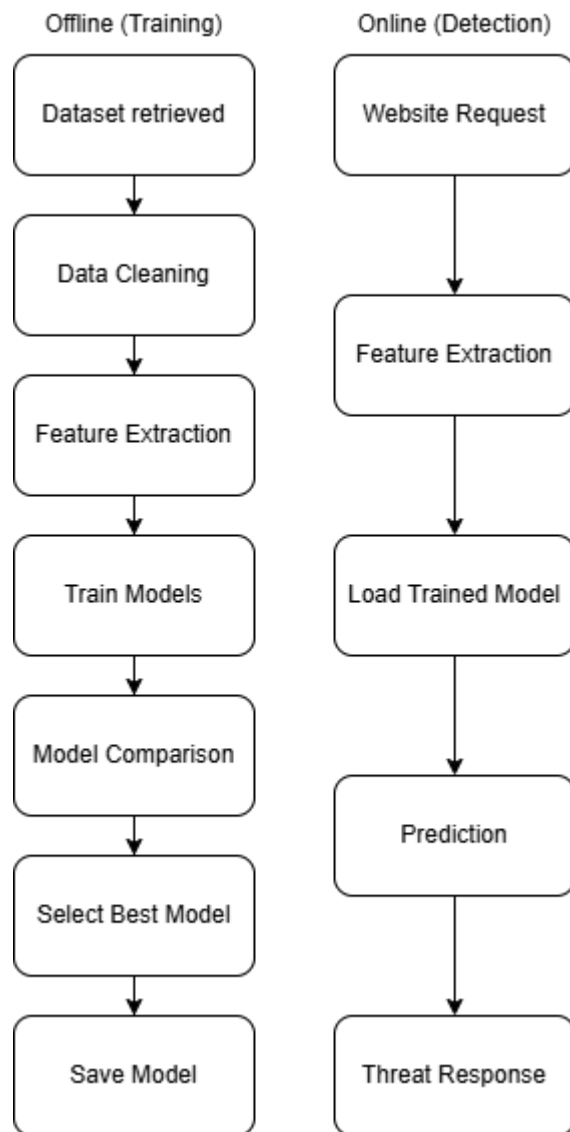


Figure 3.1 Machine Learning Pipeline

3.5 Model Evaluation

The performance of the trained models is determined using an experimental evaluation methodology. Defectiveness measures are calculated using evaluation metrics like accuracy, precision, recall, F1-score and false-positive rate.

Special concern is given to the reduction of false positives because over blocking legitimate websites can be harmful to users. The results of the evaluation are used to choose the most successful model that will be included in the final system. The chosen evaluation measures are in line with existing research on malicious web site detecting in which ensuring that false positives are minimised in deterring user confidence and system functionality are paramount.

CHAPTER 4

Implementation Plan

The implementation plan for this final year project will be carried out in multiple phases for better orientation and structure. This plan shall highlight the planned steps, tools and resources to use in phase 2 of the final year project, which should ensure the proper timeline and progress of the development, integration and evaluation of the system.

4.1 Environment Setup and Tool Preparation

The initial stage is the establishment of the development environment that will be used in the implementation of the system. Python will be adopted as the main programming language because it has a wide range of machine learning and security-related libraries. The necessary packages such as scikit-learn, pandas, NumPy, BeautifulSoup and mitmproxy will be installed and configured. Jupyter Notebook and Visual Studio Code will be used in development and testing. This stage is done to ensure the tools and dependencies are adequately ready before developing the system.

The choice of Python and its libraries is consistent with the results presented in Chapter 2, where the majority of currently existing malicious website detection systems are based on Python-based machine learning frameworks as they can be customised, have a wide library support, and can be integrated into security prototypes with ease. Even MITM proxy use has added to the real-time functionality of interception and response, which proxy-based deployments have been found to represent a good trade-off between performance and system control.

Table 4.1 Development Tools and Purpose

Tool/Framework	Purpose in System	Justification
Python	Core Development	Strong ML +Security Support
scikit-learn	ML Model Training	Proven Effectiveness
pandas/NumPy	Data Processing	Efficient Data Manipulation
BeautifulSoup	HTML Parsing	Lightweight feature extraction
mitmproxy	Traffic Interception	Real-time detection ability
Jupyter Notebook	Experimentation	Model Evaluation + Comparison

4.2 Data Gathering and Processing

During this stage, publicly accessible datasets of both legitimate and malicious websites will be gathered by working reputable sources like Kaggle and phishing intelligence archives. The data sets will be checked in terms of completeness and relevance and the redundant or conflicting entries will be deleted. Both websites will be easily marked as fraudulent or legitimate to facilitate guided learning. The assembled dataset is in turn to be split into training and testing samples to allow objective testing of models. This data preparation method is a typical indication of previous research, in which supervised learning on curated collections of phishing sites and benign websites is at the basis of effective malicious websites classifiers.

Table 4.2 Summary of Dataset Categories

Dataset Type	Source	Purpose
Malicious URLs	Kaggle	Model Training
Benign URLs	Public Website Lists/Kaggle	Baseline Comparison
HTML Content	Crawled pages	Structural Feature Extraction

4.3 Feature Extraction and Engineering

To convert raw data on the websites to meaningful numbers that can be used by machine learning models, the process of feature extraction will be done. This involves the process of deriving lexical information in the URL address, including the length of the URL, the number of special characters, and the availability of suspicious keywords. Attributes of the domain and selected webpage structural characteristics like the existence of scripts, forms or embedded objects will also be taken into account. The feature normalization and encoding techniques will be used to create consistency and to increase the performance of the models.

As outlined in the literature review, such computationally expensive features as webpage screenshots or dynamic behavioural analysis are deliberately omitted in the project. In its place, there are lightweight lexical and structural properties that are prioritised to ensure low latency and practicality to run in real-time as evidenced by numerous previous research works.

4.4 Training and Comparison of Machine Learning model

In this step, several trained machine learning classifiers will be trained on the ready dataset. Random Forest, Support Vector Machine (SVM) and Logistic Regression or Gradient Boosting are some of the models that will be evaluated and compared. Model comparison is done to make sure that the objective selection is made on empirical evidence as opposed to the choice of the algorithm. This is in line with the previous studies, in which a variety of classifiers are tested and then deployed to production.

Detection effectiveness will be measured by calculating performance measures such as accuracy, precision, recall and F1-score. The robustness will be enhanced by the use of cross-validation and hyperparameter tuning. The most effective model will be chosen to be incorporated in the detection system.

Table 4.3 Comparison of Machine Learning Models

Model	Advantages	Disadvantages	Suitability
Random Forest	High accuracy + Interpretable	Larger model size	High
SVM	Effective for high-dimensional data	Sensitive to parameters	Medium
Gradient Boosting	Very Predictive	Training time	High

4.5 Remedy Action and Threat Decision

After identifying an appropriate model, a decision mechanism based on rules will be developed to translate the results of classification into threat levels. The websites will be classified into low-risk, medium-risk, or high-risk threats based on the scores of prediction confidence. This is a threat-based model that makes the responses of the system proportional to the level of danger that is evaluated and does not affect the usability.

The decision mechanism based on threat-level is designed in a non-model driven manner to make it transparent and interpretable. This design decision complies with the industry best practises of security systems where the logic behind the decision should be presented to the users and administrators.

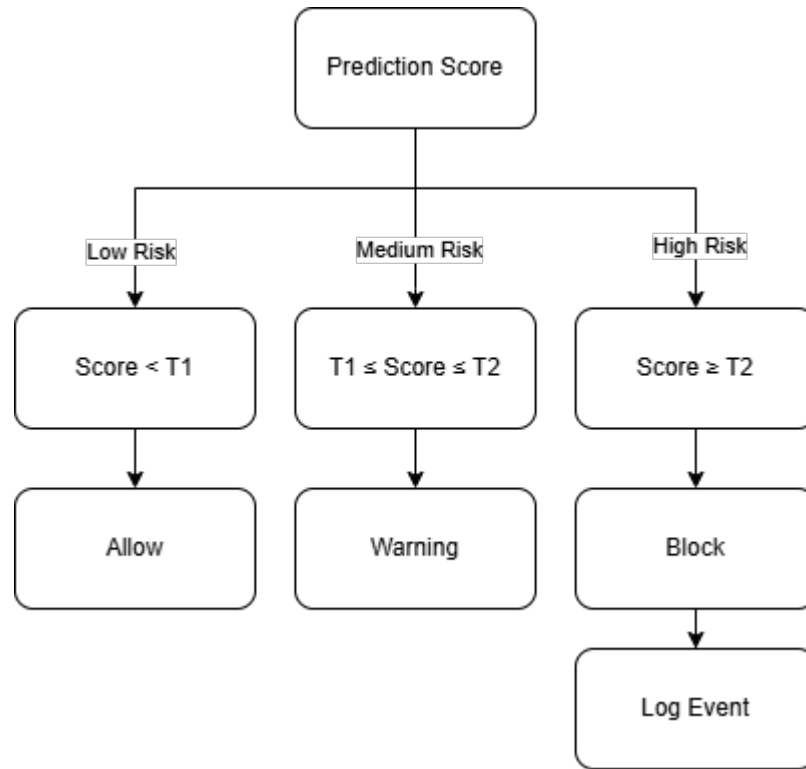


Figure 4.1 Threat Response Decision Tree

Table 4.4 Threat Levels and Corresponding Responses

Threat Level	Description	Response
Low	Legitimate/Minimal Risk	Allow Access
Medium	Suspicious behaviour	Warning Message Pop-up
High	Confirmed harmful/malicious	Block access and log

4.6 mitmproxy System Integration Phase

The chosen machine learning model will be incorporated into a local proxy-based system through the use of mitmproxy framework. mitmproxy will decipher the browser requests to the websites and send the information to the detection engine, including URLs and webpage content. The detection engine will categorize every request and provide assessment as to threat back to the proxy. This integration allows real time analysis even without the need to make changes to the browser itself.

The use of the proxy-based architecture allows to perform centralised inspection and reaction without altering browser internals. As it has been explained in Chapter 2, proxy-based systems offer a balance between the detection capacity and practicality of deployment.

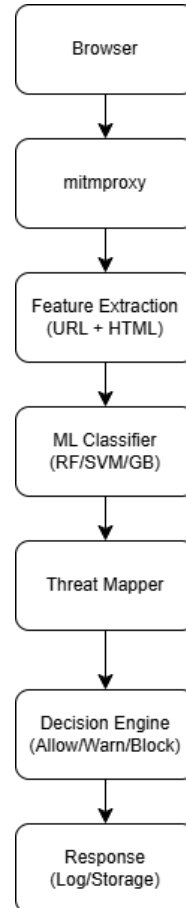


Figure 4.2 Integration Architecture

4.7 Implementation of Response Mechanism

This stage will see the implementation of response measures depending on the level of threat which is allocated. Websites with a legitimate or low level of risk will be permitted to load as usual whereas medium risk websites will invoke warning messages to warn the users. The blocking will be done on high-risk websites and all information will be logged to be analysed. The warning and blocking notifications will be displayed in simple HTML pages to guarantee the clarity and understanding on the part of the user. The response mechanism provides proportionate security measures

which minimises user disruption and provides effective protection against verified malicious websites.

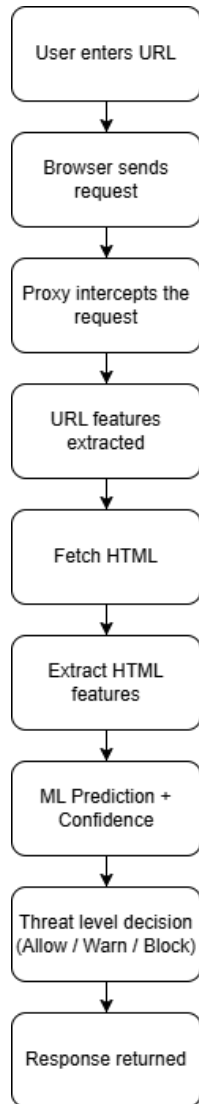


Figure 4.3 Data Flow and Response

4.8 System Testing and Evaluation

Tests will be done on the completed system with unseen samples of web sites in a controlled environment. Testing will be done on accuracy in detection, false-positive and false-negative, correctness of responses, and latency of the system. The results of the evaluation will be analysed to determine the reliability of the system and downplay the areas that will need improvement. This stage gives empirical data on system functionality.

Table 4.5 System Evaluation Metrics

Metrics	Purpose
Accuracy	Overall correctness
Precision	False positive control
Recall	Malicious site detection
F1-score	Balanced Evaluation
Latency	Real-time Feasibility

4.9 Final Reporting and Documentation

During the implementation process, the system design decisions, experimental results and testing outcomes will be logged. There will be screenshots, diagrams, and performance tables that will help to support the final report and project demonstration. The FYP 2 final thesis and presentation will be based on this documentation. Any documentation created in the implementation will provide traceability among design decisions, experimental outcomes and conclusions made on the literature review.

CHAPTER 5

CONCLUSION

5.1 Summary

The malicious and phishing websites have become a significant attacker in contemporary cybersecurity incidents and have been addressed through this project. A comprehensive literature analysis has identified that the conventional blacklist-based and signature-based detection systems could not be used to keep up with the fast adapting and obfuscated web-based attacks. As a result, the machine learning-based solutions have become a potentially viable alternative as they can be trained to generalise on the patterns and identify threats that were not observed before.

During this stage of the project, the detailed problem specification was generated based on the thorough consideration of the literature concerning the topic of feature engineering, machine learning methods, adversarial robustness, and strategies of real-time implementation. Lightweight feature extraction, supervised machine learning model comparison, and a proxy-based deployment architecture were selected as the key points of the structured methodology, based on the results of the current research. The system design and implementation plan was well designed to be feasible, interpretable and real time responsive yet at the same time be within undergraduate level of development.

5.2 Key Contributions and Justifications

The most important contribution of this project is that it was designed to develop an integrated detection and response structure, which includes malicious websites classification, and automatic security functions. Compared to most of the existing literature, which measures the accuracy of detection only, the project will specifically consider threat-level assessment and response strategies that allow to put into practise such mitigation strategies as warnings and blocking.

The literature is very potent in influencing the design decisions that were made in this project. Light weight URL and webpage structural features were chosen to strike a balance between detection capabilities and computational capabilities. Simple deep learning techniques were not the focus as they were perceived to be not as transparent, explainable, and applicable to real-time application compared to traditional supervised machine learning models. Besides, the proxy-based architecture ensures that web traffic is intercepted and analysed in real time without the need to modify the browser, thus making the system flexible and easy to deploy.

All these decisions aim to solve the gaps found in the current research, especially the absence of real-world, deployable, interpretable and practical malicious websites detection systems that combine both response and detection.

5.3 Future Works and Development

The following stage of this project (FYP 2) will be devoted to the implementation of the suggested design and the system verification by the methods of experimental evaluation. This involves the preparation of datasets, extraction of features, training and comparison of machine learning models, integrating the system using a proxy-based framework and evaluation of the system using real world sample websites. The additional improvements can be made, such as enhancing resistance to adversarial manipulation, response strategy improvement, and expansion of the system to include other threat intelligence sources.

To sum up, this project has managed to put a strong foundation on the creation of a machine learning-based malicious websites detection and responding system. The analysis design and implementation plan that has been completed illustrates the viability and applicability of the suggested solution which gives a clear and structured direction on how complete system realisation and evaluation will be made on the next project stage.

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APPENDICES

Appendix A: Meeting Logs 1-5



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 2/12/2025	MEETING NO.: 1
PROJECT ID: T88J235	
PROJECT TITLE: MACHINE LEARNING-BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM	
TRIMESTER: 2530	SUPERVISOR: HENG SWEE HUAY
STUDENT ID & Name: 1211108623 CHONG YOU XIN	CO- SUPERVISOR: -

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] - Introduction (Overview, Problem Statement, Project Objectives, Project Scope) - Literature Review
2. WORK TO BE DONE - Update Project Objectives - Continue with Chapter 3: Methodology
3. PROBLEMS ENCOUNTERED Change the report into UK spelling.
4. COMMENTS Please proceed.


 Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)


 Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MEETING DATE: 26/12/2025	MEETING NO.: 2
PROJECT ID: T88J235	
PROJECT TITLE: MACHINE LEARNING-BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM	
TRIMESTER: 2530	SUPERVISOR: HENG SWEE HUAY
STUDENT ID & Name: 1211108623 CHONG YOU XIN	CO- SUPERVISOR: -

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Methodology 3.1 - 3.5
2. WORK TO BE DONE Continue with Chapter 4 (Currently considering 'Implementation' for this chapter. I plan to talk about what systems I plan to use, how to develop, etc.)
3. PROBLEMS ENCOUNTERED - How will the meeting log for the library literature meeting work? Should we write a separate log for it or will it be provided to us? - Are there any suggestions or amendments to my current progress?
4. COMMENTS Please proceed.


Supervisor's Signature &
Stamp


Co-Supervisor's Signature
& Stamp (if any)


Student's Signature

NOTES:

- Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
- For FYP Phase 1, total six log sheets are to be submitted (every other week*).
- For FYP Phase 2, total six log sheets are to be submitted (every other week**).
- Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MEETING DATE: 12/1/2026	MEETING NO.: 3
PROJECT ID: T88J235	
PROJECT TITLE: MACHINE LEARNING-BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM	
TRIMESTER: 2530	SUPERVISOR: HENG SWEE HUAY
STUDENT ID & Name: 1211108623 CHONG YOU XIN	CO- SUPERVISOR: -

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Implementation Plan (Only texts)
2. WORK TO BE DONE - Add graphs/tables, etc. - Finalising Chapter 1
3. PROBLEMS ENCOUNTERED If in phase 2, things don't go as planned in the implementation plan, am I allowed to make amends in the final report and the project?
4. COMMENTS Yes, you may still make changes in Phase 2. Please work harder.


 Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)


 Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MEETING DATE: 17/1/2026	MEETING NO.: 4
PROJECT ID: T88J235	
PROJECT TITLE: MACHINE LEARNING-BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM	
TRIMESTER: 2530	SUPERVISOR: HENG SWEE HUAY
STUDENT ID & Name: 1211108623 CHONG YOU XIN	CO- SUPERVISOR: -

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] - Amended Chapter 1: Literature Review - Added Abstract + Acknowledgement + Conclusion - Completed Chapter 4: Implementation Plan	2. WORK TO BE DONE - Finalising/Checking Report - Make any necessary amendments
3. PROBLEMS ENCOUNTERED Just for confirmation, the plagiarism/similarity rate mentioned in the guidelines is below 10%. Are there any limitations to AI rate?	
4. COMMENTS Please improve accordingly.	

 _____ Supervisor's Signature & Stamp	_____ Co-Supervisor's Signature & Stamp (if any)	 _____ Student's Signature
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NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

MEETING DATE: 19/1/2026	MEETING NO.: 5
PROJECT ID: T88J235	
PROJECT TITLE: MACHINE LEARNING-BASED MALICIOUS WEBSITE DETECTION AND RESPONSE SYSTEM	
TRIMESTER: 2530	SUPERVISOR: HENG SWEE HUAY
STUDENT ID & Name: 1211108623 CHONG YOU XIN	CO- SUPERVISOR: -

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] - Added figures and tables - Finalised report
2. WORK TO BE DONE
3. PROBLEMS ENCOUNTERED Is there anything else I need to amend/include before submitting the report?
4. COMMENTS Please wrap up the report.


 Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)


 Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Appendix B: LLC Attendance List

FYP for FIST: Attendance List

Session 2: 10/12/2025

The following students have attended the FYP session (FIST) with the library.

Trainer:


MELISSA DE VALDA BINTI MOHD YATIM
Assistant Manager, Library
Information Services & Research Support Unit
Malay Open University
Jalan Ayer Keroh Lama
75450 Marang

Melissa De Valda Binti Mohd Yatim
Librarian, Tun Dr. Hasmah Mohd Ali Digital Library

No.	Full Name	MMU ID No.	FYP Title
1	Wai Yu Heng	1221101101	YU-GI-OH! INFORMATION SHARING SYSTEM
2	CHONG YOU XIN	1211108623	Machine learning based malicious website detection and response system
3	Ng Xin Hui	1211107961	AI-Driven Dashboard for Crowdsourced Forest Fire Detection and Prediction
4	LYE JING HUEY	1211109625	Students' Perception of ChatGPT: A Study Among Malaysian University Students
5	IVAN TEE HONG WEI	1211105898	GAIT RECOGNITION WITH TRANSFER LEARNING USING CONVOLUTIONAL NEURAL NETWORK
6	Sek Joo Shan	1211108716	Autonomous drone navigation using Computer Vision
7	Ganeesh Kumar	1211100996	LIGHTWEIGHT VULNERABILITY DETECTION PLATFORM USING OPEN-SOURCE TOOLS
8	Ho Yi En	241UT2410J	Computer-Vision Based Object Detection Models in Traffic-based Scenarios
9	CHIN JET NING	1231301358	Context-Awareness Shopping Assistant
10	FONG QUAN MING	243UT246WQ	SPEECH EMOTION DETECTION
11	Wong Yan Yee	1211111866	Few shot learning emotion recognition using gait analysis
12	Anderson Tan Kuan Lu	12213050909	Malicious URL Detection Using AI
13	Terrance Teh Teng Hong	1231302616	chatbot for university examination guidelines
14	Lai Wen Jun	1211108712	Tourism Data Analytics For Tourism Behaviour and Market Segmentation
15	NICHOLAS TAY JUN YANG	242UT24371	AI FRAMEWORK FOR FINANCIAL TABLE UNDERSTANDING
16	Lim Chun Hao	1211109659	ScratchJr Modules With E-Portfolio
17	KUEH PANG TENG	1211112304	ALZHEIMER'S DISEASE PREDICTION USING MRI DATA
18	SIOW WAN SOON	1221101294	METaverse-READY IOT SMART HOME INTERFACE FOR ENERGY AND COMFORT CONTROL
19	KING YIN BEN	1211201970	Intelligent vision inspection system for pcb defect detection using ai
20	Quah Yu Wei	1211110756	Development Of A Provably secure E-Voting Scheme
21	Soo Teck Shen	1211108213	Intelligent Agriculture Data Analytics And Monitoring Platform
22	KO XUE NING	242UT244BS	Smart Budget Calculator Website
23	SAMUEL ONG LI KANG	243UT2452R	IoT & Edge for Decentralized Water Quality Assurance with Blockchain
24	Teh Wei Yun	1211112194	ARTIFICIAL INTELLIGENCE IN SUSTAINABILITY MARKETING
25	Gerald Prakash Joseph	1211100598	IOT-BASED INDOOR ENVIRONMENTAL AND OCCUPANT SAFETY MONITORING SYSTEM
26	JASON CHU JUN HON	1211111951	COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS FOR WASTE CLASSIFICATION
27	Jamie Lim Shi Ting	242UT2449F	WEB-BASED PET GROOMING BOOKING SYSTEM
28	WONG JIAN LIN	241UT240N3	Game Revenue Prediction and Business Insight Dashboard using Power BI

Appendix C: Checklist for Interim Report Submission



Checklist for Interim Report Submission

(To be filled in by Student)

STUDENT'S DETAILS

Project Code	FIST
Name	CHONG YOU XIN
ID No	1211108623
Title of Thesis	Machine Learning Based Malicious Website Detection and Response System
Supervisor Name	Heng Swee Huay

REPORT ARRANGEMENT	√	Comments (if any differences)
1. Cover of The Interim Report	√	
2. Title Page of the Interim Report	√	
3 Copyright page of I Interim Report	√	
4. Declaration Page of Interim report	√	
5. Acknowledgement	√	
6. Table of Contents	√	
7. Abstract	√	
8. List of Tables	√	
9. List of Figures	√	
10. List of Symbols	√	
11. List of Appendices	√	
12. Chapter 1: Introduction – objectives, scope	√	
13. Chapter 2: Literature Review	√	
14. Chapter 3: Methodology	√	
15. Chapter 4: Implementation Plan	√	
16. Chapter 5: Conclusion	√	
17. References – APA style	√	
18. Appendices	√	
19. CD/ DVD and envelope as shown in Appendix K		N/A, Softcopy Only.
20. Attachment: FYP Meeting Logs (all) 1 set	√	

FORMAT OF REPORT	√	Comments
1. Page Numbering	√	
2. Font and Type Face	√	
3. Font Cover	√	
4. Tables and Figures	√	
5. Comb Bind		N/A
6. Colour of the Front Cover		N/A
7. Number of words > 5000 (Main content only)	√	

Checked by

A handwritten signature in black ink, appearing to be the Chinese characters '张' (Zhang) followed by a stylized flourish.

Student's Signature & Date
CHONG YOU XIN, 21.01.2026